



1717 Wakonade Drive
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October 17, 2022

L-PI-22-042
10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Prairie Island Nuclear Generating Plant Units 1 and 2
Docket Nos. 50-282 and 50-306
Renewed Facility Operating License Nos. DPR-42 and DPR-60

Prairie Island Nuclear Generating Plant Licensee Event Report 2022-001-00

Pursuant to 10 CFR 50.73(a)(2)(i)(B), Northern States Power Company, a Minnesota corporation, doing business as Xcel Energy (hereafter "NSPM"), hereby submits Prairie Island Nuclear Generating Plant (PINGP) Licensee Event Report (LER) 50-282/2022-001-00.

Summary of Commitments

This letter makes no new commitments and no revisions to existing commitments.

A handwritten signature in black ink, appearing to read 'Chris Domingos', written over a horizontal line.

Christopher P. Domingos
Site Vice President, Monticello and Prairie Island Nuclear Generating Plants
Northern States Power Company – Minnesota

Enclosure

cc: Administrator, Region III, USNRC
Project Manager, Prairie Island, USNRC
Resident Inspector, Prairie Island, USNRC
State of Minnesota

ENCLOSURE 1
PRAIRIE ISLAND NUCLEAR GENERATING PLANT
LICENSEE EVENT REPORT 50-282/2022-001-00

3 pages follow



LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form)

<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk aid: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name Prairie Island Nuclear Generating Plant, Unit 1	2. Docket Number 05000 -282	3. Page 1 OF 3
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4. Title
Fuel Assemblies incorrectly placed in Spent Fuel Pool

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
08	15	2022	22	- 001 -	00	10	17	2022	Prairie Island, Unit 2	05000 -306
									Facility Name	Docket Number
										05000

9. Operating Mode 1	10. Power Level 100
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11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	
☐ OTHER (Specify here, in abstract, or NRC 366A).				

12. Licensee Contact for this LER

Licensee Contact Nathan Fedora, Senior Nuclear Regulatory Engineer	Phone Number (Include area code) 612-342-8971
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13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS

14. Supplemental Report Expected

☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)
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15. Expected Submission Date

Month	Day	Year

On August 17, 2022, with both Unit 1 and Unit 2 of the Prairie Island Nuclear Generating Plant (PINGP) operating in Mode 1 at 100 percent power, it was discovered that 38 fuel assemblies had been placed in an incorrect location of the Spent Fuel Pool (SFP) during the move of fuel assemblies which commenced at 12:30 on August 15, 2022 and was completed on August 16, 2022. The location of 29 of these 38 fuel assemblies did not comply with Technical Specification Design Feature 4.3.1.1, resulting in Technical Specification 3.7.17 Action Statement A being entered on August 17, 2022, at 10:53. The cause of this condition was a procedural step that wasn't followed resulting in an incorrect date being entered in the software configuration tool. This incorrect date caused the software configuration tool to calculate longer fuel assembly cooling times resulting in mis-categorization of the fuel assemblies.

Following discovery, the correct date in the software configuration tool was used to ensure the non-complying fuel assemblies were moved to the correct locations to restore compliance and Technical Specification LCO 3.7.17 Action Statement A was exited on August 19, 2022, at 10:32.

This event is being reported pursuant to 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by technical specifications.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME	2. DOCKET NUMBER	YEAR	3. LER NUMBER SEQUENTIAL NUMBER	REV NO.
Prairie Island Nuclear Generating Plant	05000- -282	2022	- 001	- 00

NARRATIVE**EVENT DESCRIPTION**

On August 17, 2022, with both Unit 1 and Unit 2 of the Prairie Island Nuclear Generating Plant (PINGP) operating in Mode 1 at 100 percent power, it was discovered that 38 fuel assemblies had been placed in an incorrect location of the Spent Fuel Pool (SFP). The location of 29 of the 38 fuel assemblies did not comply with Technical Specification Design Feature 4.3.1.1 resulting in Technical Specification 3.7.17 Action Statement A being entered on August 17, 2022, at 10:53.

On August 4, 2022 Fuel Transfer Logs were generated using the software configuration tool in preparation for upcoming fuel assembly moves in anticipation of a core offload scheduled to occur during an upcoming refueling outage on Unit 1. The generation of the Fuel Transfer Logs unknowingly carried over a planning date of June 1, 2029 which had been previously used for a scoping study of future SFP configurations. This incorrect planning date resulted in the software configuration tool calculating longer fuel assembly cooling times and mis-categorizing 38 fuel assemblies such that they were assigned a less reactive category than they should have been. The fuel assemblies were classified as Fuel Category 6 when they should have been classified as Fuel Category 4 or 5.

On August 15, 2022, the move of fuel assemblies commenced at 12:30 using the Fuel Transfer Logs that were generated on August 4, 2022 with the incorrect planning date. The fuel assembly moves were completed on August 16, 2022.

On August 17, 2022, during preparations for the move of new fuel assemblies from the new fuel pit to the SFP, new Fuel Transfer Logs were generated in software configuration tool using a correct planning date. This process resulted in receipt of multiple warnings which alerted engineering of the configuration errors. Investigation into the warnings identified the incorrect planning date that had been previously used.

These fuel assembly configuration errors resulted in non-shutdown Technical Specification 3.7.17 Action Statement A being entered on August 17, 2022, at 10:53.

This event is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by technical specifications.

Human Performance

The Reactor Engineer did not use the guidance contained within the work instructions for the SFP fuel assembly verification when entering the planning date in the software configuration tool.

CAUSE OF THE EVENT

The cause of this condition was a procedural step that was not followed resulting in an incorrect planning date being allowed to remain in the software configuration tool. Contributing to this was the lack of adequate guidance for verification of critical parameters within the associated surveillance procedure that would ensure SFP configuration would conform to Technical Specification Design Feature 4.3.1.1.

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Prairie Island Nuclear Generating Plant	05000- -282	2022	- 001	- 00

NARRATIVE**ASSESSMENT OF SAFETY CONSEQUENCES**

There were no safety consequences related to this condition.

The SFP criticality analysis evaluates accident conditions in the SFP including the mis-load of multiple fresh fuel assemblies. Calculations show that the amount of soluble boron concentration in the SFP required to offset the highest reactivity increase caused by accident conditions would be 2030 ppm. The boron concentration was measured once during the time the fuel assemblies were not compliant with Technical Specification Design Feature 4.3.1.1. This measurement was 2873 ppm, which was taken on August 17, 2022 at 12:25. The boron requirements for keff remained met even with the 29 fuel assemblies loaded in the incorrect location. Therefore, the SFP never entered an unanalyzed condition and did not exceed any safety limits.

This event does not represent a safety system functional failure for Unit 1 or Unit 2. There were no radiological, environmental, or industrial impacts associated with this event. The health and safety of the public and site personnel were not impacted during this event.

CORRECTIVE ACTIONS

The planning date was corrected in the software configuration tool and action was taken to move the 29 non-complying fuel assemblies to acceptable locations. This action restored compliance with Technical Specification Design Feature 4.3.1.1 and allowed Technical Specification 3.7.17 Action Statement A to be exited on August 19, 2022, at 10:32.

To enhance procedural guidance, procedure changes will be made to the relevant procedures to require explicit verification of critical parameters including the planning date, discharge date, burnup, and enrichment.

PREVIOUS SIMILAR EVENTS

No previous similar events have occurred at PINGP in the prior 3 years.

ENERGY INDUSTRY IDENTIFICATION SYSTEM (EIS) CODES

Not applicable to this report, as this is not against an equipment / component issue.